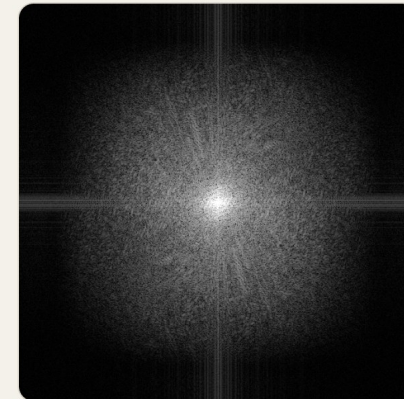
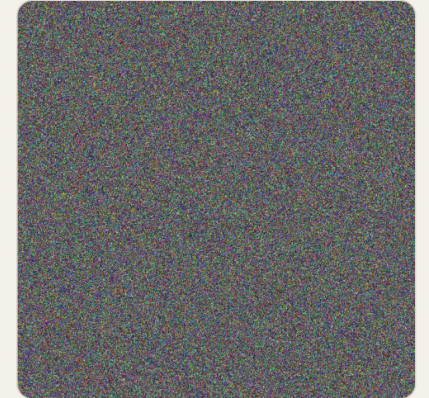


PROJECT PRESENTATION

PhaseForge

Encrypting, watermarking, filtering and cleaning up images and audio in the frequency domain.

Team PhaseForge • Mahir Mahtab and Saffone Ahmed
CSE 220 • 27 September 2026



Problem and goal

The problem

Fourier-domain methods for image security, watermarking and speech clean-up are usually seen only as equations. It is hard to see what they do to real data, or where they fail.

Our goal

One application that applies these methods to real images and audio, shows every result, and tests how robust they are.

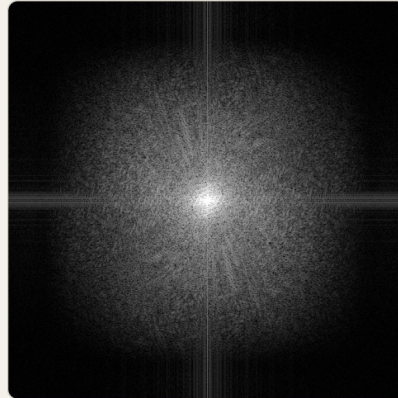
THEORY

Magnitude and phase

The Fourier transform splits a signal into waves. Magnitude says how strong each wave is; phase says where it sits, and carries the structure.



photo A



A's magnitude
spectrum

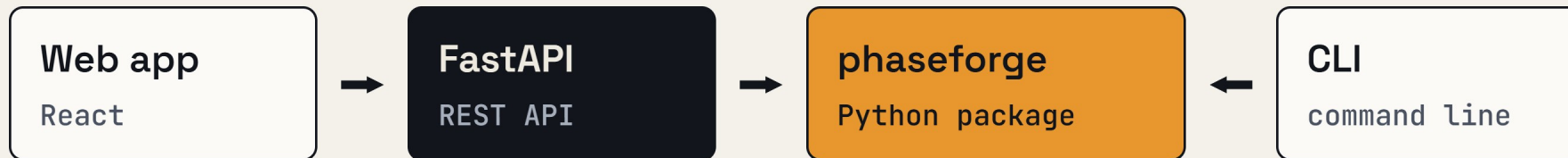


photo B



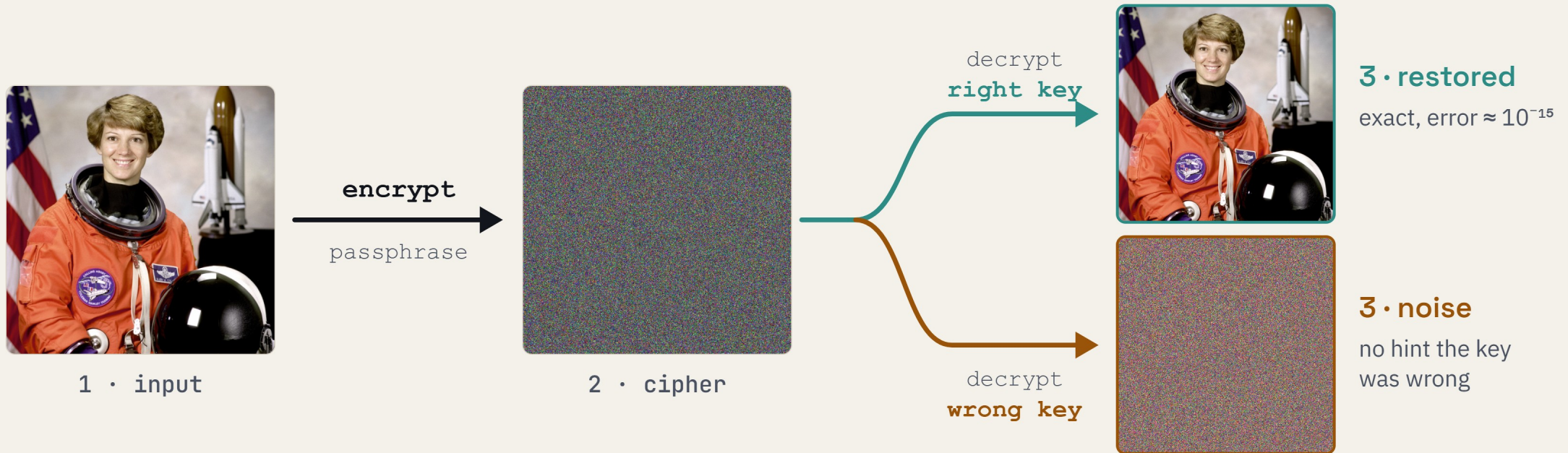
A's phase + B's
magnitude

Transform, modify, transform back



Double random phase encoding

encrypt = $\times$ mask A $\rightarrow$ FFT $\rightarrow$ $\times$ mask B $\rightarrow$ inverse FFT • masks come from the passphrase



Key PBKDF2, 200 000 rounds, random salt • Audio same method, block by block

Damaged ciphers still decrypt

Damage to the encrypted file spreads as light noise over the whole image, instead of destroying part of it.



original



5 % noise • 31.8 dB



20 % noise • 19.8 dB

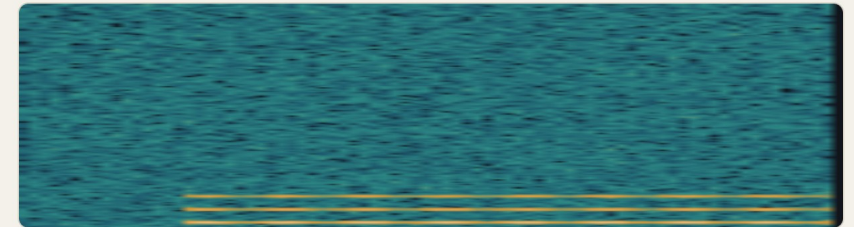


10 % lost • 17.7 dB

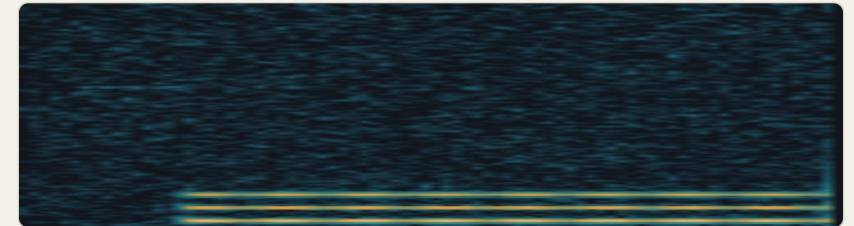
Denoise and Enhance



Denoise	OM-LSA: turns down each bin by how likely it is noise
Enhance	adds harmonic repair, speech-band EQ and level
Result	19 dB less noise, speech level unchanged



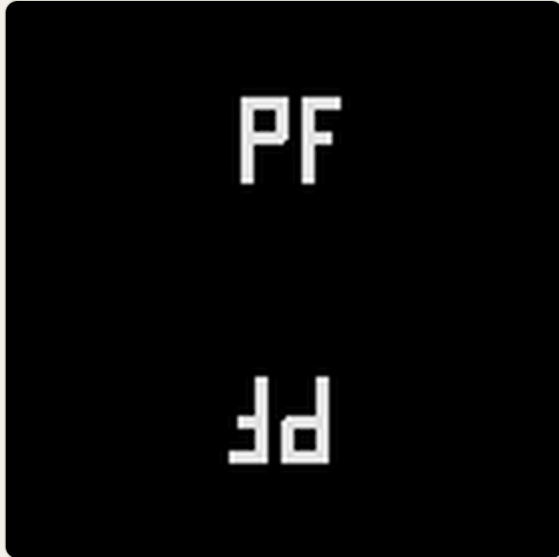
before



after

OTHER FEATURES

Watermark • Filter



Watermark

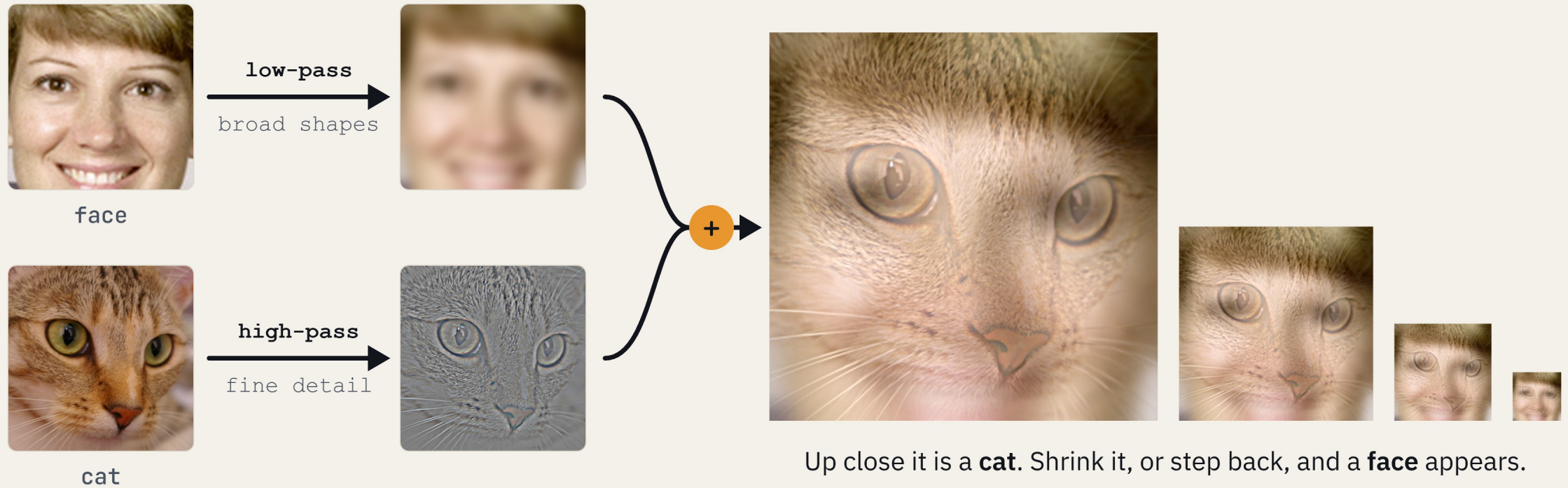
hides a mark in the magnitude spectrum;
invisible, recoverable



Filter

low-, high- or band-pass with Gaussian,
Butterworth or ideal masks

One picture, two images



The eye sees fine detail only up close; from far away only the broad shapes remain.

LIVE DEMO

PhaseForge in use

Encrypt and decrypt · watermark · denoise

THANK YOU

Questions?

Team PhaseForge • Mahir Mahtab and Saffone Ahmed